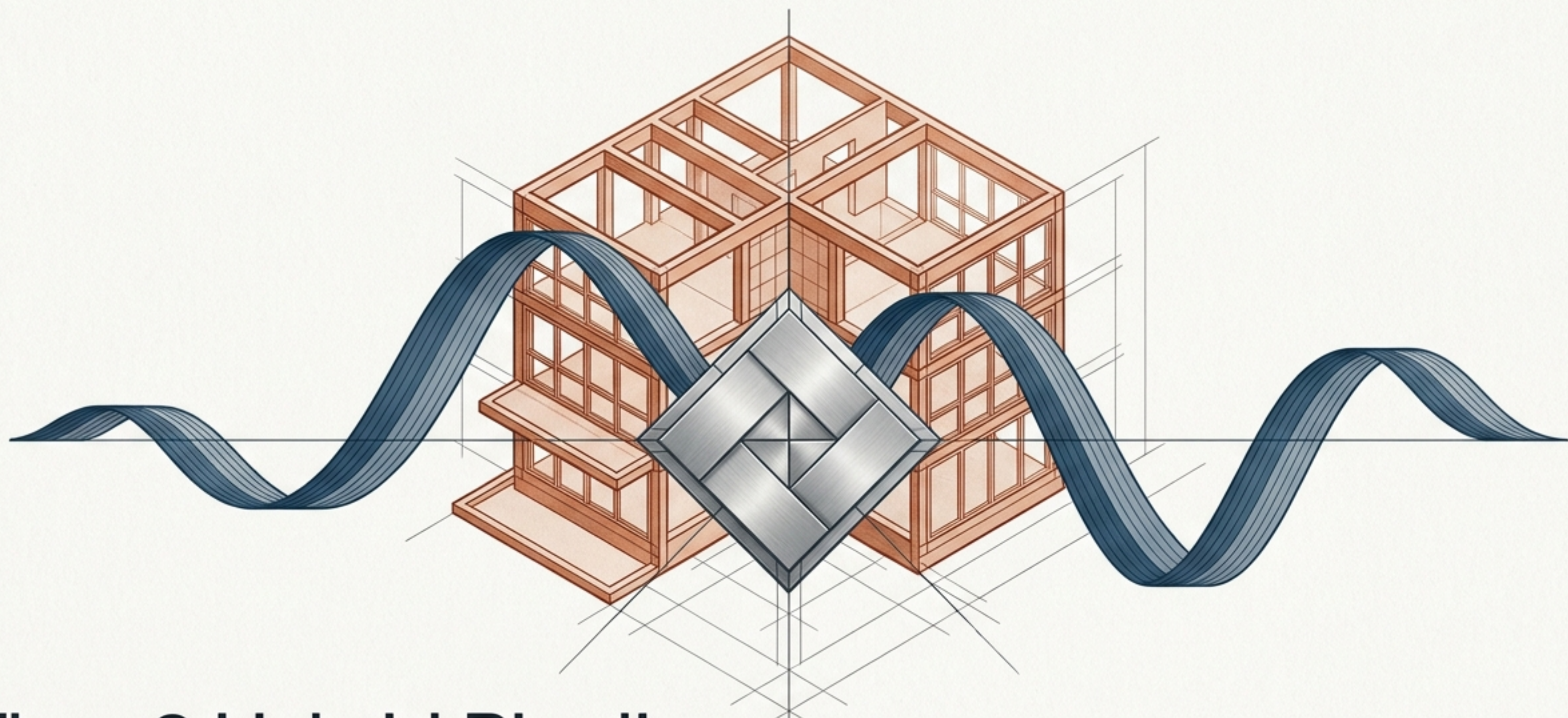
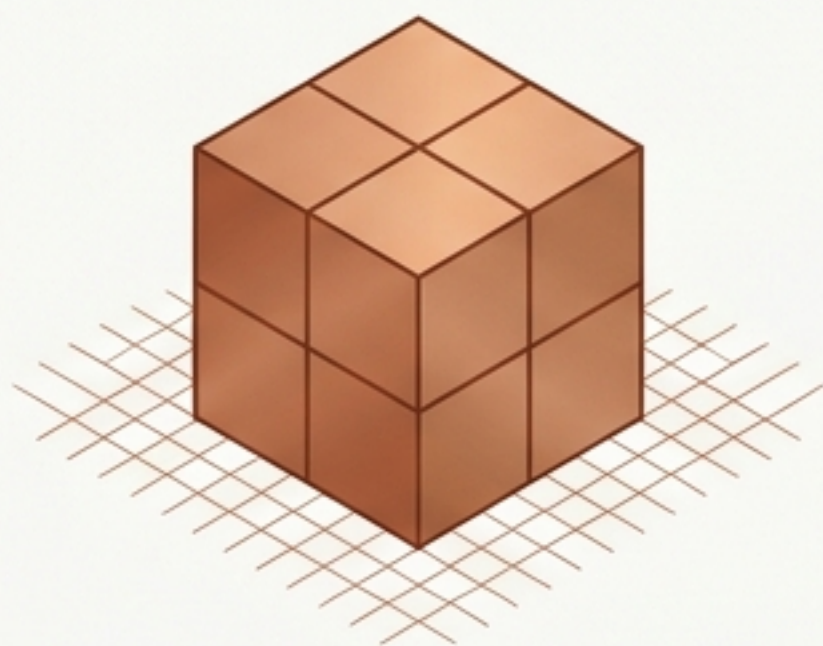




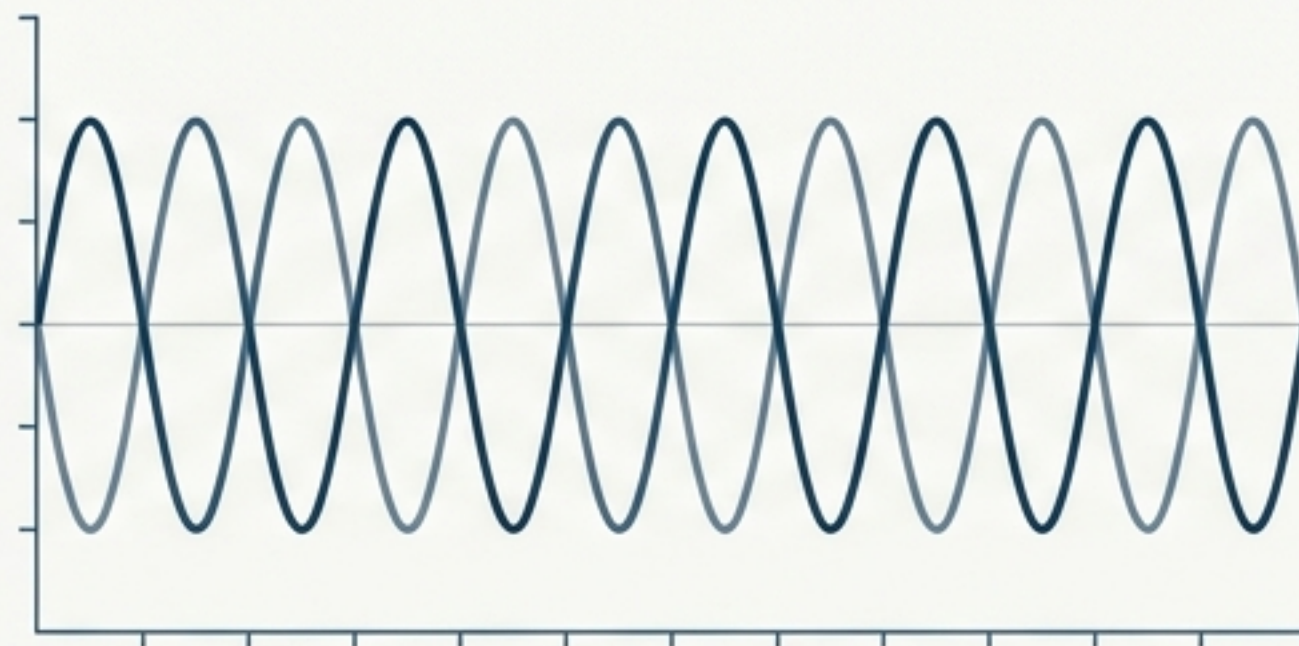
The v3 Hybrid Pipeline

Integrating spatial features with time-evolving market dynamics via LSTM + MLP

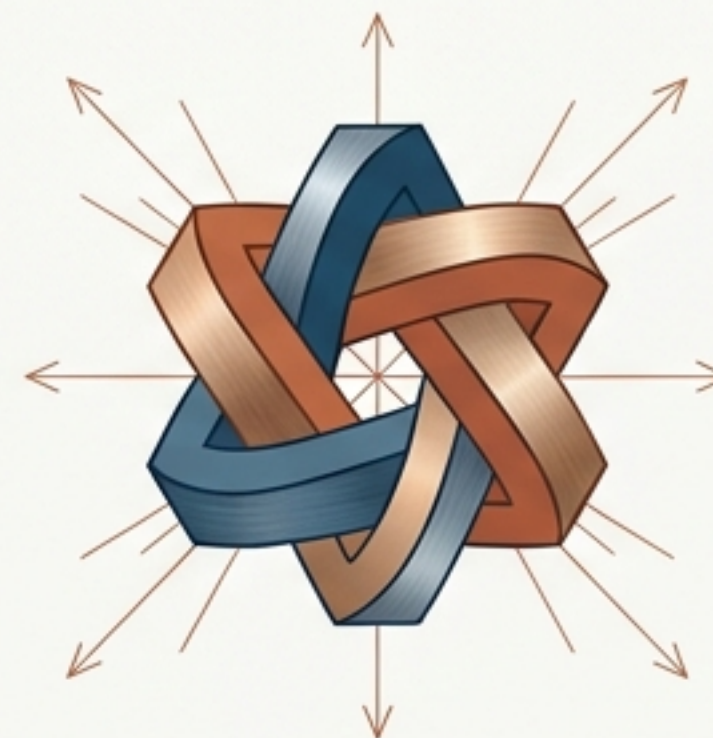
Beyond Tabular Rows



Static Property Specs



12-Month Market Timeline



Precision Pricing

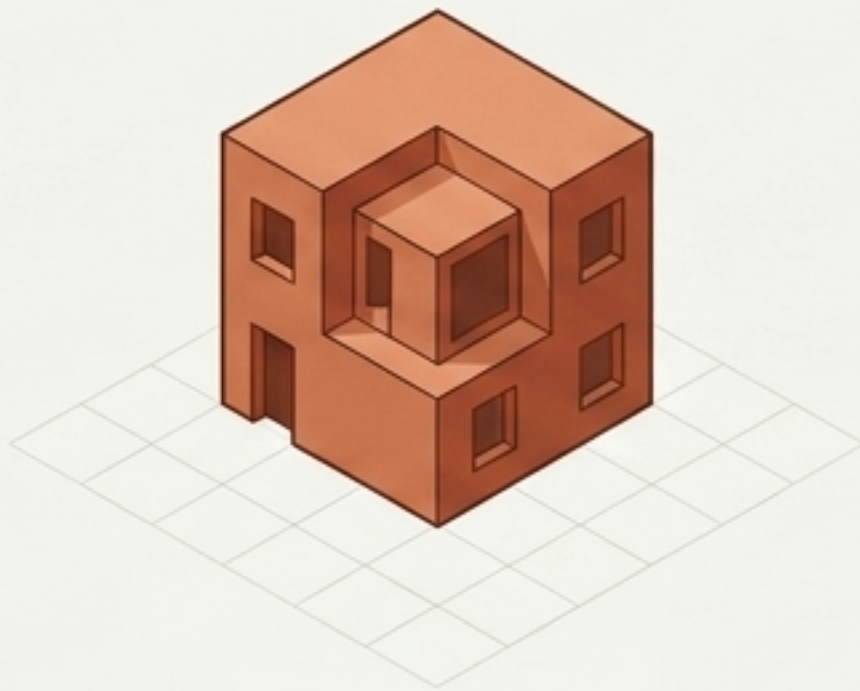
Moving away from purely per-row tabular analysis, the v3 architecture natively understands market heating and cooling.

By deploying a dual-branch neural network, it unites physical property attributes with shifting segment-level trends to predict resale prices with unprecedented temporal awareness.

The Blindspot of Gradient-Boosted Trees

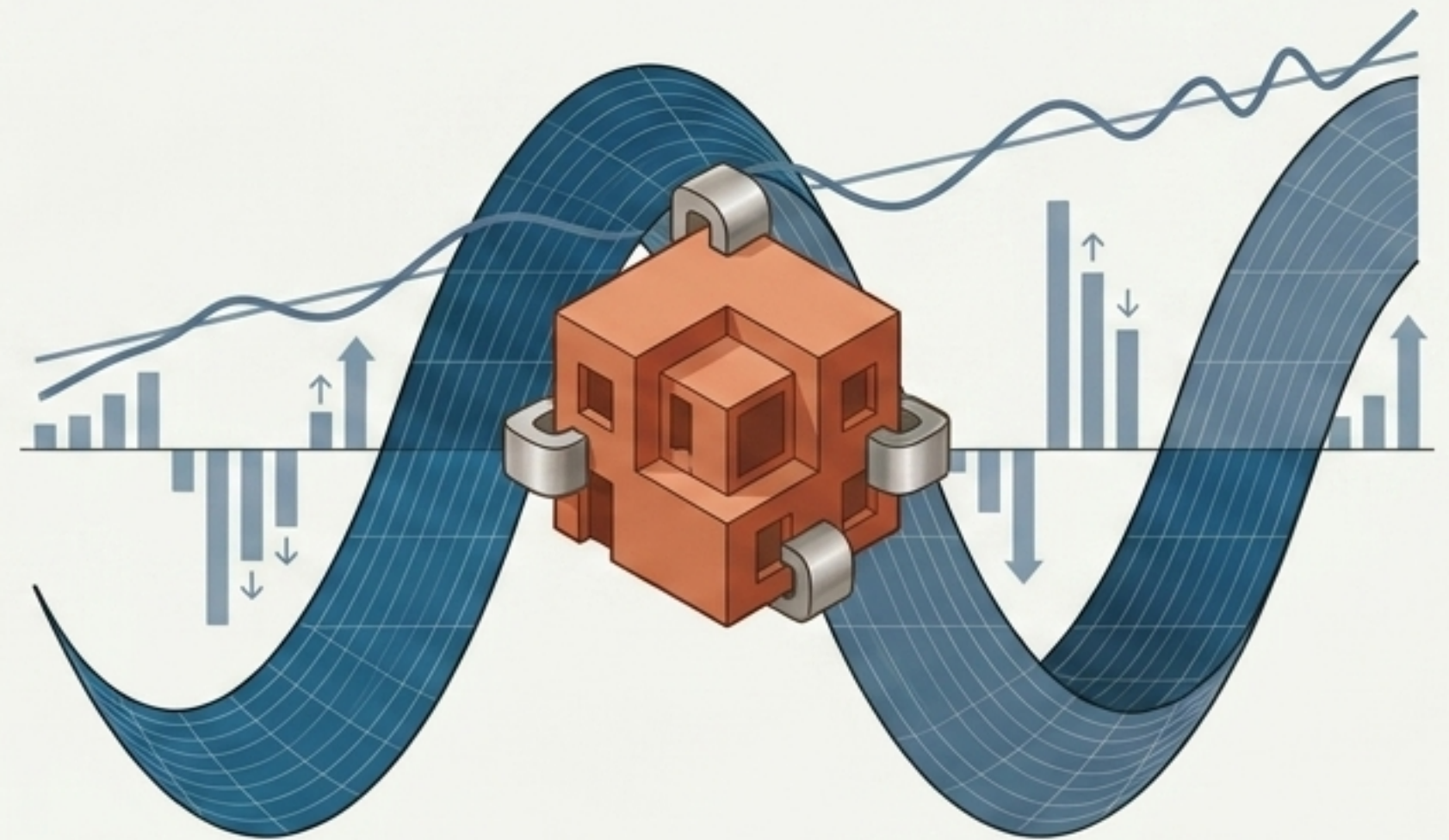
Resale rows are tabular, but every transaction sits within a time-evolving market. Without explicit temporal structures, tree-based models (v2/v4) are blind to month-to-month segment-level price shifts, volume changes, and macro drift (RPI).

What Trees See



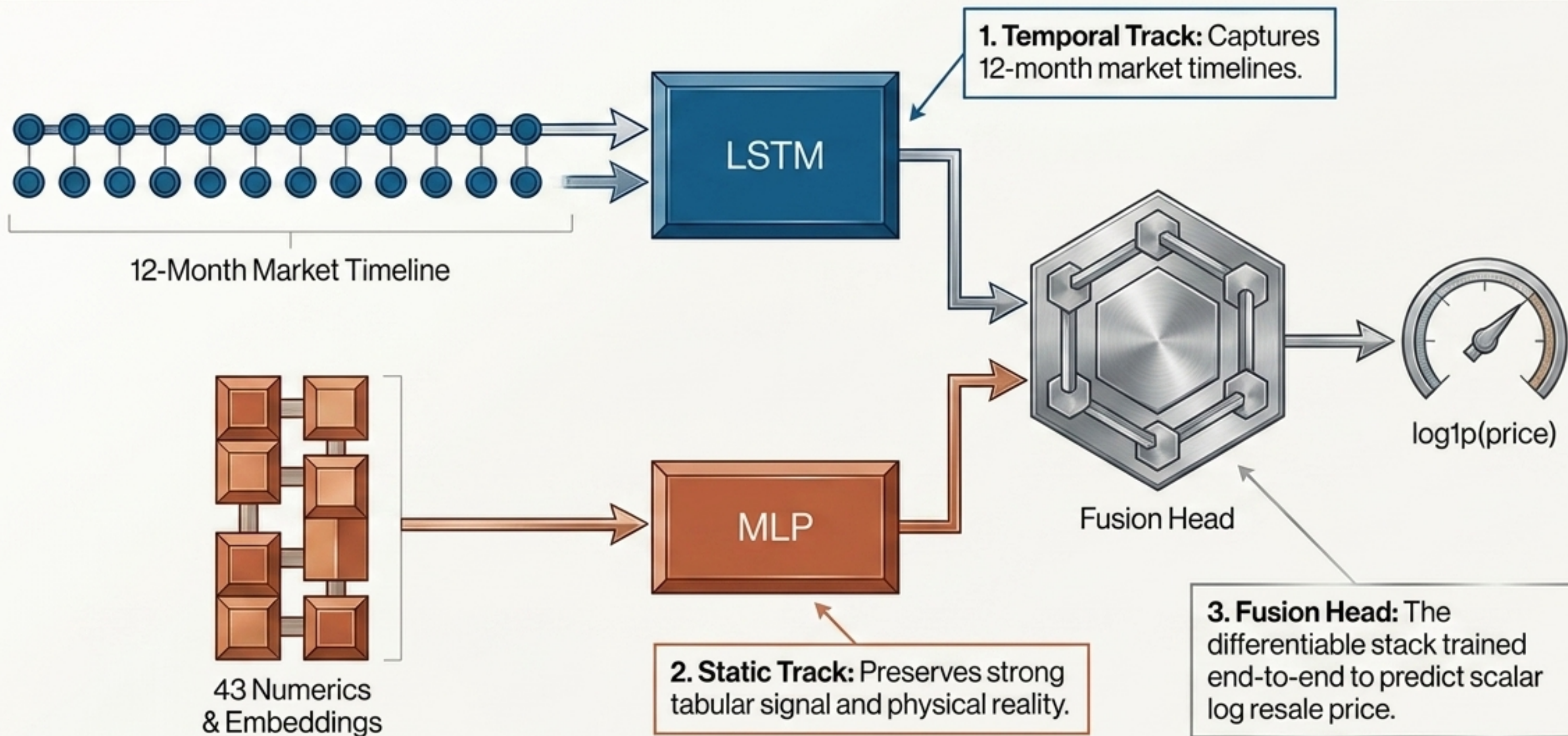
Static feature interactions treated in strict isolation.

What Reality Is



Embedded in a continuously shifting economic macro-environment.

The Dual-Track Architecture



Encoding Market Dynamics

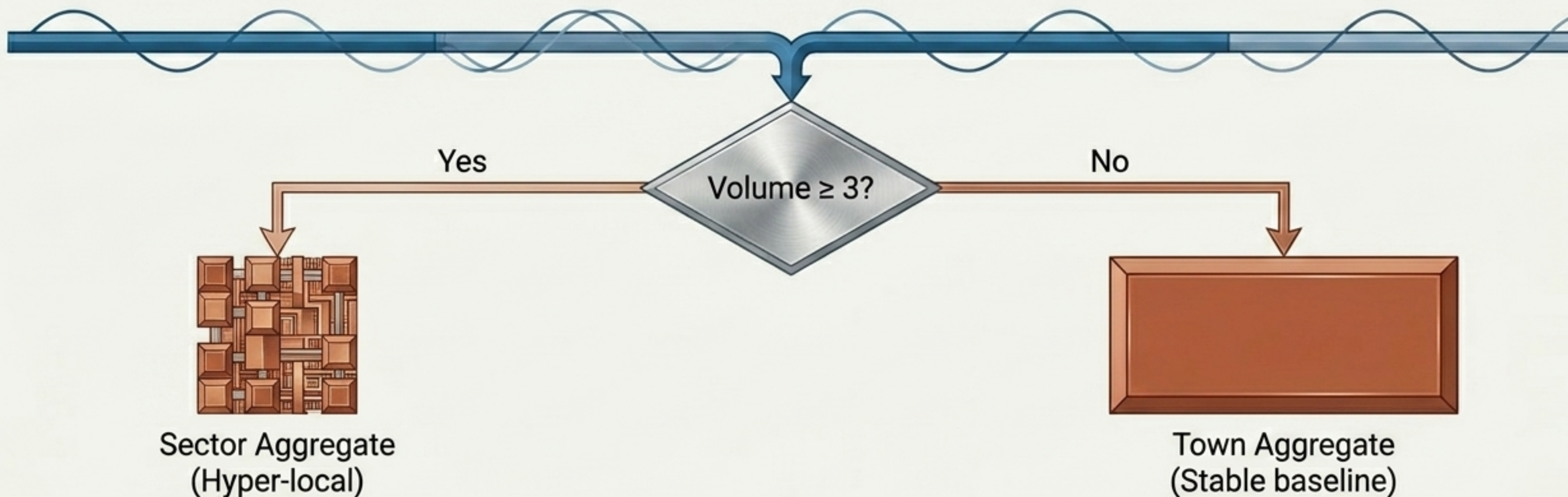
The Temporal Prior acts as a short-panel sequence representing 12 calendar months of local market history. It acts as a learned summary of trend, volatility, and seasonality.



log_mean_price	Segment typical log price	
log_std_price	Dispersion of log prices	
log_volume	Liquidity (log1p transaction count)	
log_mean_area	Composition shift	
hdb_rpi	Official RPI (one-quarter lag)	
month_sin	Cyclical calendar encoding	
month_cos	Cyclical calendar encoding	

Balancing Granularity and Noise

To prevent the LSTM from training on low-volume noise, the pipeline uses a safer hybrid aggregation logic. The default timeline is based on town. If a specific postal sector experiences dense enough transaction volume, that localized data is substituted in. Sparse or missing codes automatically drop back to the stable town baseline.

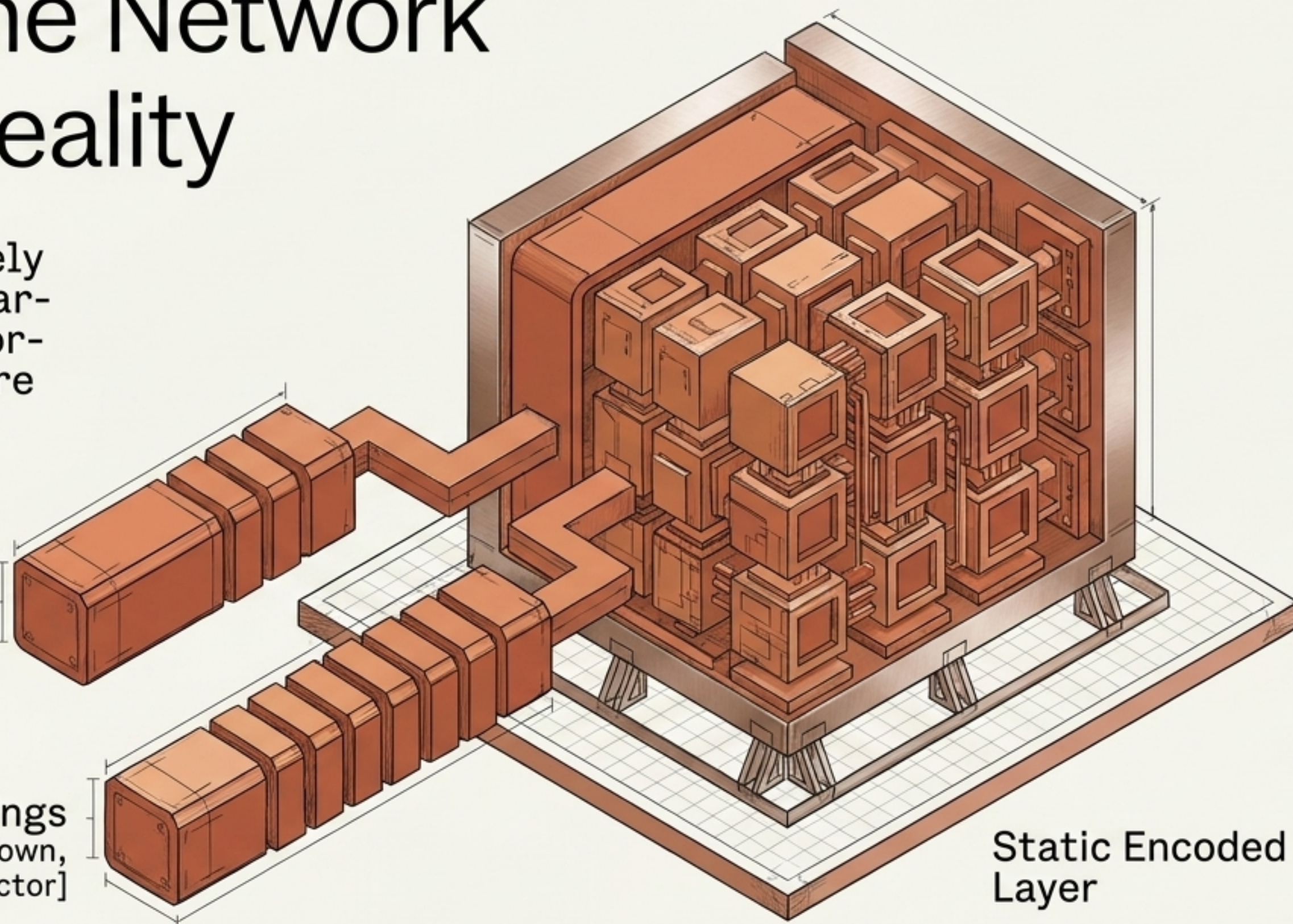


Grounding the Network in Physical Reality

The architecture is not purely temporal. The static path carries the vital unit-level information, matching the feature philosophy of the v2 model.

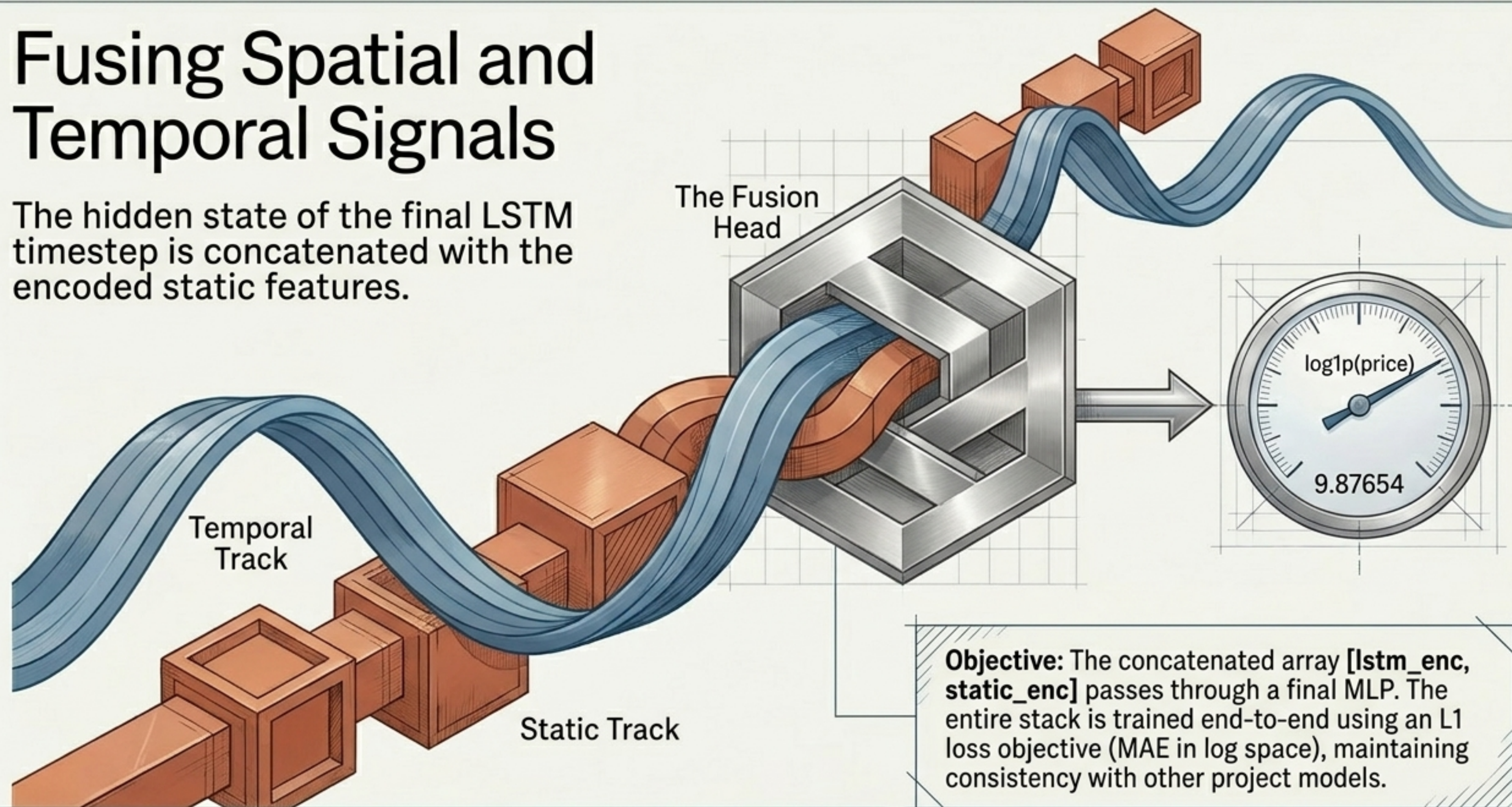
43 Scaled Numerics
[Area, Remaining Lease,
Storey, KDTree Spatial Encodings]

Categorical Embeddings
[flat_type, flat_model, town,
MRT/school names, postal_sector]

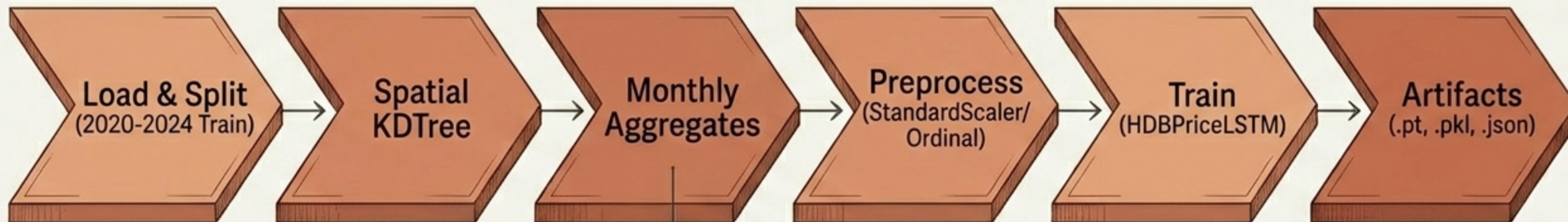


Fusing Spatial and Temporal Signals

The hidden state of the final LSTM timestep is concatenated with the encoded static features.



The Execution Pipeline



Strict Leakage Prevention

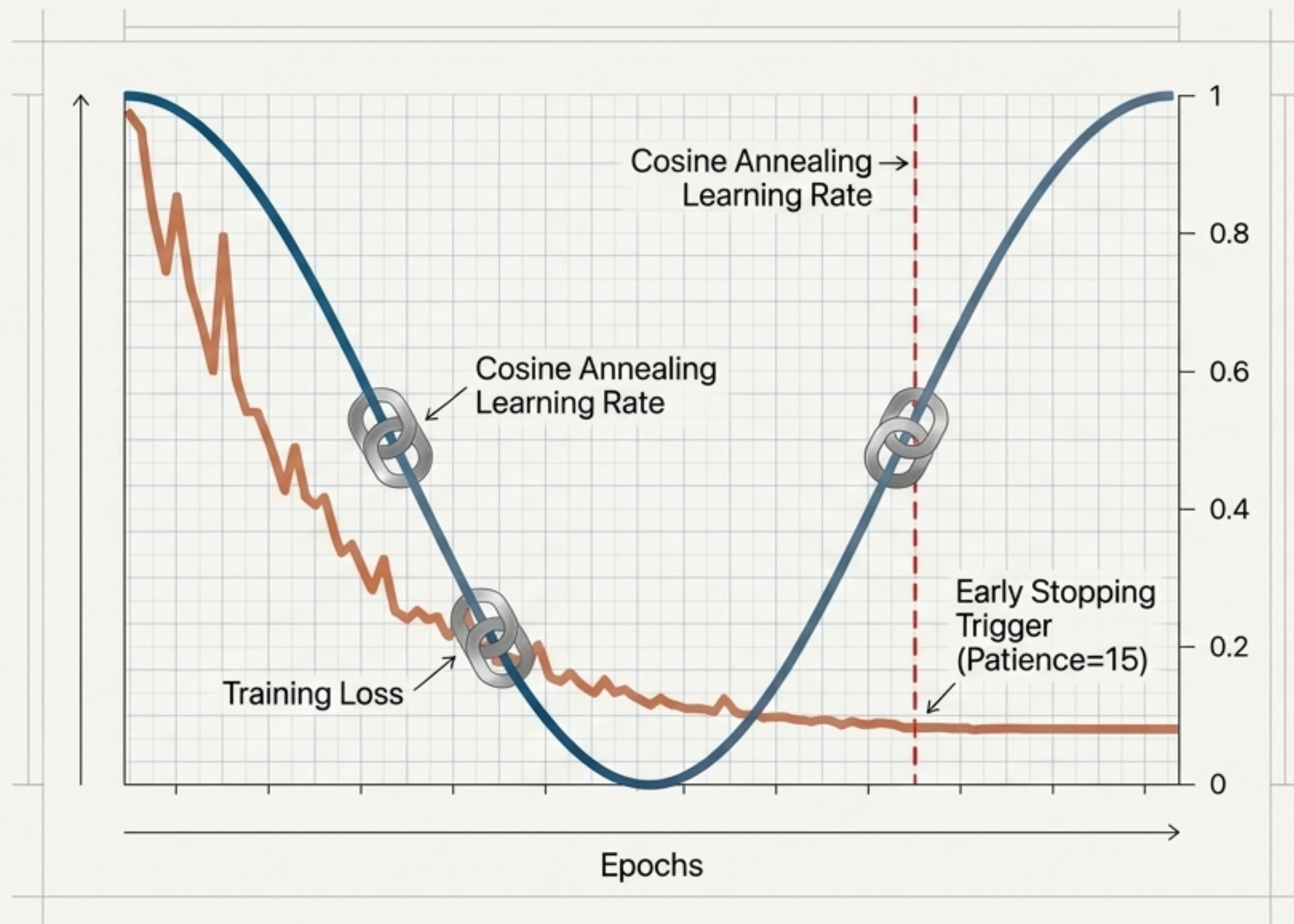
Monthly aggregations enforce strict temporal rules. The `build_monthly_agg` function ensures models only use data from months strictly before each row's transaction month ($< \text{curr_p}$). Widening sequences without this rule destroys model validity.

Network Optimization and Stability

Optimizer: AdamW with gradient clipping.

Learning Rate: Cosine annealing decay toward $\eta_{\min}$.

Convergence: Early stopping triggered if validation loss fails to improve for 15 consecutive epochs. The best weights are saved from the lowest validation loss epoch.



Interpreting the Metrics



Evaluation reveals a natural forward-year step up.

Train predictably outperforms Val and Test due to the forward-year gap, though R^2 remains exceptionally high.

The Unshuffled Fix

Older versions evaluated train predictions using `shuffle=True`, permuting predictions against labels and generating nonsense train metrics.

v3 enforces `train_eval_loader(..., shuffle=False)` for accurate historical tracking.

Architectural Diagnostic

Trees (v2/v4)

Hybrid LSTM (v3)

Temporal
Context

Engineered interactions only

Explicit 12-month
calendar spine

Core Signal

Per-row tabular

Town/Sector segments +
unit tabular

Compute
Load

Lower compute, CPU viable

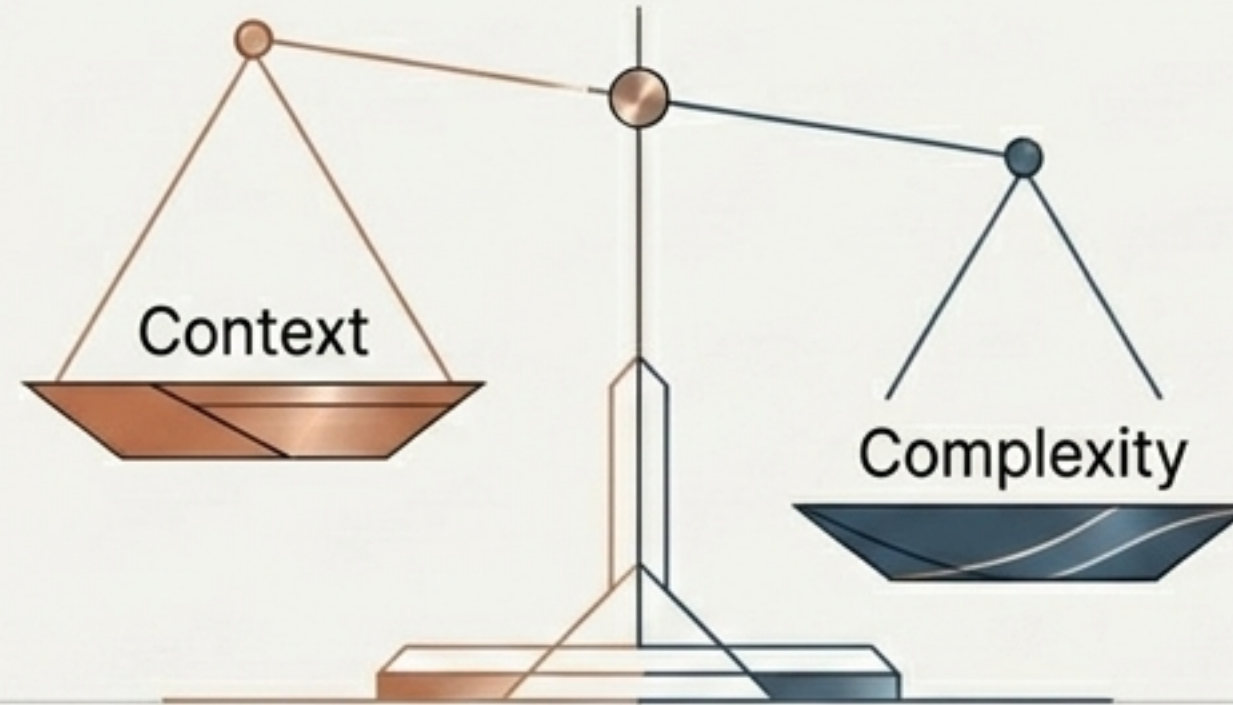
High complexity, PyTorch
GPU required

Leakage
Risk

Low

High risk without strict
pre-transaction rules

Evaluating the Trade-Offs



Pros

- Aligns perfectly with how practitioners think about market heating/cooling.
- Keeps strong tabular signals while providing a fully differentiable stack.
- Input sequences are interpretable aggregated statistics.

Cons

- Computationally heavy, requiring PyTorch GPU environment.
- Lower global feature interpretability compared to tree partial dependence.
- High risk of distribution shift if the 2026 market behaves radically differently than training-era sequences, degrading the temporal prior.

Production Viability



The v3 hybrid model successfully bridges the gap between static unit evaluation and macro-economic timing. Because absolute MAE/MAPE vary by initialization, do not evaluate v3 on a single lucky epoch. Final deployment viability must be determined by directly comparing Val + Test performance against v4 (CatBoost/ARIMA) on identical data splits.